
FAILURE TRACE F
cases that failed under h
from the session log
CANDIDATE h'
proposer edits rho, G, K, M
inside edit budget b_t
G1 SCOPE
$\Delta(h, h') \subseteq \mathcal{P}_{gen} \ \wedge \ |\Delta| \le b_t$
$O(|\Delta|)$
no model runs
pass
G2 LEAK
$\neg \exists v \in V : \ v \sqsubseteq \text{add}(\Delta)$
$O(|\Delta| + |V|)$
Aho-Corasick
pass
G3 TRIGGER
$\overline{\hat{p}(h', x) - \hat{p}(h, x)}_F > \overline{\delta(x)}_F$
$O(2k|F|\,C_{run})$
live runs
pass
G4 PRESERVE
$|\{x \in P : \hat{p}(h, x) - \hat{p}(h', x) > \delta(x)\}| \le m$
$O(2k|P|\,C_{run})$
live runs
pass
G5 NOISEGATE
$\bar{s}(h') \ge \bar{s}(h) - \tau$
$O(2k|T|\,C_{run})$
verdict only
pass
G6 COST
$\Delta s > \tau \Rightarrow \Delta c \le \kappa \Delta s$; else $\Delta c < 0 \vee h' \subset h$
$O(1)$
uses G5 runs
pass
G7 TRIPWIRE
$E_{eval} \cap (\text{Canary} \cup \text{Checker} \cup \text{Ref}) = \emptyset$
$O(|E_{eval}|)$
event scan
pass
G8 ROLLBACK
$\text{H}(h) \in S \ \wedge \ S \cap W_{agent} = \emptyset$
$O(1)$
store lookup
reject on first failure
all eight pass
OUTCOME A ADMIT
version stored by H(h') in S
admission record in gate log
keyed by (model, executor)
OUTCOME B VARIANT ARCHIVE
h' kept with failed check
and the cases where it helped
proposer sees check id only